

User → Website Widget → Backend → AI → Avatar → Response



Database Save



Notification (WhatsApp/SMS)



Client Dashboard

1. Super Admin
all data showing
2. Client (Business owner)
only leads show
Status update (New / Contacted / Booked)
3. Visitor
only ai talk (no login)

Admin and database

1. Dashboard

Total Clients
Active Clients
Total conversations
Leads Today / This Week
Recent Activity (latest chats)

Users Table => id, email, password_hash, role, client_id, created_at

2. Client Management

List of Clients

clients Table => id, business_name, logo_url, primary_colour, heygen_avatar_id, greeting_message, embed_api_key, status, notification_whatsapp, notification_sms, notification_email, created_at

Add New Client

Form fields:

- Business Name
- Logo upload

- Primary Color
 - email
 - Avatar ID
 - Message
 - Status
 - Notification settings:
 - WhatsApp number
 - SMS number
 - Email
-

Edit Client

- Update all details
- Change avatar
- Change notification settings
- Suspend / Activate client

3. Knowledge Base: client data save and ai answer

Uploaded files (PDF / DOCX)

Added URLs

Text data

Knowledge_sources Table=> id, client_id, source_type
(pdf|url|docx|text), original_filename_or_url, ingested_at

Form fields :

- **Type** → pdf / url / text / docx
- **Source** → file name / URL / short text preview
- **Created Date**
- **Actions:**
 - Delete

4. Knowledge_chunks:

Knowledge_chunks Table=> id, client_id, source_id, content, embedding (vector), created_at

5. All Conversation:

Admin can see ALL client conversations data showing

Table: id, client_id, session_id, started_at, ended_at, visitor_name, visitor_email, visitor_phone, transcript (jsonb)/full chat (user or ai), summary, suggested_action, urgency_flag, status (new|contacted|booked|lost),owner_notes

6. Messages Table=> id, conversation_id, role, content, timestamp

7. Audit_log Table => id, user_id, action, metadata, timestamp

8. Notification Settings:

- WhatsApp alerts :Enable / Disable channels
- SMS alerts
- Email alerts

Admin Panel

|

|— Dashboard

|— Clients

| |— Add Client

| |— Edit Client

| |— View Client

|— Knowledge Base

|— Conversations

|— Embed Codes

|— Notifications

|— Logs

|— Settings

You are a senior full-stack SaaS architect and AI systems engineer.

Build a production-ready MVP called "SpeakingSites".

STACK:

- Frontend: Next.js 15 + Tailwind + TypeScript
- Backend: FastAPI (Python)
- Database/Auth/Vector DB: Supabase + pgvector
- AI Model: Anthropic Claude
- STT: OpenAI Whisper
- Avatar Streaming: HeyGen LiveAvatar Lite API
- Notifications: Twilio + Resend
- Hosting: Vercel + Railway
- Widget: Vanilla JS embeddable script

GOAL:

Build a multi-tenant AI avatar platform where businesses embed an AI receptionist/avatar on their website.

CORE REQUIREMENTS:

1. Super Admin dashboard
2. Client dashboard
3. Knowledge base ingestion + RAG
4. Live avatar conversation system
5. Lead capture
6. Conversation storage
7. Follow-up tracking
8. WhatsApp/SMS/email notifications
9. Multi-tenant isolation
10. Production-grade reliability

ARCHITECTURE:

- Browser never accesses Claude/OpenAI/HeyGen directly
- All AI APIs remain server-side
- Widget communicates only with our backend
- Use stateless sessions
- Use silent retries
- Use graceful degradation to text chat if WebRTC fails

DATABASE TABLES:

(users, clients, knowledge_sources, knowledge_chunks, conversations, messages, audit_log)

IMPLEMENT:

- Supabase Row Level Security
- JWT auth
- Embeddable widget script
- Conversation transcript storage
- AI-generated summaries
- Suggested follow-up actions
- Urgency detection
- Contact extraction from transcripts

KNOWLEDGE BASE PIPELINE:

- Upload PDF/DOCX/TXT
- Scrape URLs
- Extract text
- Chunk into 500-token chunks
- Generate embeddings
- Store in pgvector
- Retrieve top-k chunks during conversations

WIDGET FLOW:

1. Load widget.js
2. Fetch config via API key
3. Render launcher
4. Create session
5. Open HeyGen LiveAvatar
6. Stream audio/text
7. Process with Claude + RAG
8. Return avatar response
9. Save transcript on end
10. Trigger notifications

CLIENT DASHBOARD:

- Recent conversations
- Status tracking

- Notes
- Follow-up queue
- Transcript viewer

ADMIN DASHBOARD:

- Create/edit/suspend clients
- Upload training materials
- Generate embed codes
- View all conversations

NON-NEGOTIABLE RELIABILITY:

- Stateless sessions
- Graceful fallback
- Connection health checks
- Silent retries
- Never expose stack traces to users

OUTPUT:

- Full folder structure
- SQL schema
- API routes
- Supabase policies
- FastAPI services
- Widget architecture
- RAG pipeline
- WebSocket architecture
- Deployment instructions
- Environment variables
- MVP-ready production code

IMPORTANT:

Optimize for:

- Fast MVP shipping
- Single-developer maintainability
- Low operational complexity
- Easy onboarding
- Multi-tenant security
- Real-world production use

Use clean architecture and modular services.